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Scheduled scaling for Amazon EC2 Auto Scaling

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With scheduled scaling, you can set up automatic scaling for your application based on predictable load changes. You create scheduled actions that increase or decrease your group's desired capacity at specific times.

For example, you experience a regular weekly traffic pattern where load increases midweek and declines toward the end of the week. You can configure a scaling schedule in Amazon EC2 Auto Scaling that aligns with this pattern:

- On Wednesday morning, one scheduled action increases capacity by increasing the previously set desired capacity of the Auto Scaling group.
- On Friday evening, another scheduled action decreases capacity by decreasing the previously set desired capacity of the Auto Scaling group.

These scheduled scaling actions allow you to optimize costs and performance. Your application has sufficient capacity to handle the midweek traffic peak, but does not over-

provision unneeded capacity at other times.

You can use scheduled scaling and scaling policies together to get the benefits of both approaches to scaling. After a scheduled scaling action runs, the scaling policy can continue to make decisions about whether to further scale capacity. This helps you ensure that you have sufficient capacity to handle the load for your application. While your application scales to match demand, current capacity must fall within the minimum and maximum capacity that was set by your scheduled action.

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How scheduled scaling works

To use scheduled scaling, create *scheduled actions*, which tell Amazon EC2 Auto Scaling to perform scaling activities at specific times. When you create a scheduled action, you specify the Auto Scaling group, when the scaling activity should occur, the new desired capacity, and optionally a new minimum capacity and a new maximum capacity. You can create scheduled actions that scale one time only or that scale on a recurring schedule.

At the specified time, Amazon EC2 Auto Scaling scales based on the new capacity values, by comparing current capacity to the specified desired capacity.

- If current capacity is less than the specified desired capacity, Amazon EC2 Auto Scaling scales out, or adds instances, to the specified desired capacity.
- If current capacity is greater than the specified desired capacity, Amazon EC2 Auto Scaling scales in, or removes instances, to the specified desired capacity.

A scheduled action sets the group's desired, minimum, and maximum capacity at the date and time specified. You can create a scheduled action for only one of these capacities at a time, for example, desired capacity. However, there are some cases where

you must include the minimum and maximum capacity to ensure that the desired capacity that you specified in the action is not outside of these limits.

Recurring schedules

To create a recurring schedule using the AWS CLI or an SDK, specify a cron expression and a time zone to describe when that scheduled action is to recur. You can optionally specify a date and time for the start time, the end time, or both.

To create a recurring schedule using the AWS Management Console, specify the recurrence pattern, time zone, start time, and optional end time of your scheduled action. All of the recurrence pattern options are based on cron expressions. Alternatively, you can write your own custom cron expression.

The supported cron expression format consists of five fields separated by white spaces: [Minute] [Hour] [Day_of_Month] [Month_of_Year] [Day_of_Week]. For example, the cron expression `30 6 * * 2` configures a scheduled action that recurs every Tuesday at 6:30 AM. The asterisk is used as a wildcard to match all values for a field. For other examples of cron expressions, see <https://crontab.guru/examples.html>  (<https://crontab.guru/examples.html>). For information about writing your own cron expressions in this format, see [Crontab](http://crontab.org)  (<http://crontab.org>).

Choose your start and end times carefully. Keep the following in mind:

- If you specify a start time, Amazon EC2 Auto Scaling performs the action at this time, and then performs the action based on the specified recurrence.
- If you specify an end time, the action stops repeating after this time. A scheduled action does not persist in your account once it has reached its end time.
- If a recurrence time exactly matches the end time, Amazon EC2 Auto Scaling will not perform the scheduled action at the end time.
- The start time and end time must be set in UTC when you use the AWS CLI or an SDK.

Time zone

By default, the recurring schedules that you set are in Coordinated Universal Time (UTC). You can change the time zone to correspond to your local time zone or a time zone for

another part of your network. When you specify a time zone that observes Daylight Saving Time (DST), the action automatically adjusts for DST.

The valid values are the canonical names for time zones from the Internet Assigned Numbers Authority (IANA) Time Zone database. For example, US Eastern time is canonically identified as `America/New_York`. For more information, see <https://www.iana.org/time-zones>  (<https://www.iana.org/time-zones>).

Location-based time zones such as `America/New_York` automatically adjust for DST. However, a UTC-based time zone such as `Etc/UTC` is an absolute time and will not adjust for DST.

For example, you have a recurring schedule whose time zone is `America/New_York`. The first scaling action happens in the `America/New_York` time zone before DST starts. The next scaling action happens in the `America/New_York` time zone after DST starts. The first action starts at 8:00 AM UTC-5 in local time, while the second time starts at 8:00 AM UTC-4 in local time.

If you create a scheduled action using the AWS Management Console and specify a time zone that observes DST, both the recurring schedule and the start and end times automatically adjust for DST.

Considerations

When you create a scheduled action, keep the following in mind:

- The order of execution for scheduled actions is guaranteed within the same group, but not for scheduled actions across groups.
- A scheduled action generally runs within seconds. However, the action might be delayed for up to two minutes from the scheduled start time. Because scheduled actions within an Auto Scaling group are executed in the order that they are specified, actions with scheduled start times close to each other can take longer to execute.
- You can temporarily turn off scheduled scaling for an Auto Scaling group by suspending the `ScheduledActions` process. This helps you prevent scheduled actions from being active without having to delete them. You can then resume scheduled scaling when you want to use it again. For more information, see [Suspend and resume Amazon EC2 Auto Scaling processes \(./as-suspend-resume-processes.html\)](#).

- After creating a scheduled action, you can update any of its settings except the name.
- When multiple scheduled actions within the same Auto Scaling group have identical cron expressions, the execution order becomes arbitrary and undefined. To ensure predictable behavior, you can use unique scheduled start times for each scheduled action.

Limitations

- The names of scheduled actions must be unique per Auto Scaling group.
- A scheduled action must have a unique time value. If you attempt to schedule an activity at a time when another scaling activity is already scheduled, the call is rejected and returns an error indicating that a scheduled action with this scheduled start time already exists.
- You can create a maximum of 125 scheduled actions per Auto Scaling group.

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Developer Tools

[AWS Code Example Library](https://docs.aws.amazon.com/code-library/latest/ug/what-is-code-library.html)

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[AWS CLI](https://docs.aws.amazon.com/cli/) (<https://docs.aws.amazon.com/cli/>)

Helpful Links

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